CE205 Databases and Information Retrieval

Performance Measures for Evaluation of a Machine Learning based Intelligent System and calculation of RFM scores for customers

Class 5

True Positives (TP): When a sample of the **positive** class is also detected as **positive** by the system.

False Negative (FN): When a sample of the **positive** class is detected as **negative** by the system.

True Negative (TN): When a sample of the **negative** class is also detected as a **negative** by the system.

False Positive (FP): When a sample of the **negative** class is detected as a **positive** by the system.

Performance Measures:

One of the metrics is precision (PR); it is a ratio between true positives and predicted positives. Imbalance classes have a stronger effect on the computation of precision. The relation for PR is given by Eq. (1a). The other commonly used metric is recall (RC). It is a ratio between true positives and actual positives; the formula is given by Eq. (1b).

$$PR = \frac{TP}{TP + FP} \qquad (1a)$$

$$RC = \frac{TN}{TN + FP} \qquad (1b)$$

Accuracy and F1 score provide the overall performance of the classification of instances. The decision about an instance should not be made if the values of these metrics are low. In a two-class problem, there is a 50% chance (when classes are balanced) that the instance will belong to a group1 or group2. If the accuracy of classification is less than 50% then it is just a random guess and it is not suitable to take any decision. The accuracy (ACC) is calculated using the relation given in Eq. (2).

$$ACC = \frac{TP + TN}{TP + FN + TN + FP} \qquad (2)$$

F1 score can be computed using Eq. 3. Like accuracy, the data distribution also affects the F1 score, and therefore, is not a reliable choice for imbalanced data.

$$F1\ \text{score} = 2 \times \frac{PR \times RC}{PR + RC} \qquad (3)$$

In real-life problems, the distribution of the classes is normally imbalanced. For instance, class1 is 70% and class2 is 30%, if a machine learning algorithm is not trained well and it predicts all samples as a class1, the obtained accuracy will be 70%. Still, this high accuracy is meaningless. One of the possible solutions is balanced accuracy (BACC), which is computed using Eq. 4.

$$BACC = \frac{1}{2}(SEN + SPE) \qquad (4)$$

where *SEN* and *SPE* represent sensitivity and specificity, respectively, and given by Eq. 7 and 9.

Some other metrics to observe the performance of the classification for imbalanced classes are geometric mean (GM) and Youden's index (YI). These metrics are defined by Eq. 5 and 6, respectively.

$$GM = \sqrt{SEN \times SPE} \quad (5)$$

$$YI = SEN + SEP - 1 \quad (6)$$

Two metrics are given by Eq. 7 and 8, sensitivity (SEN) and Predicted Positive Condition Rate (PPRC) provide accurately detected true positives and predicted true positives, respectively. The specificity (SPE) is given by Eq. (9). It determines accurately detected true negatives.

$$SEN = \frac{TP}{TP + FN} \quad (7)$$

$$PPCR = \frac{TP + FP}{TP + FN + TN + FP} \quad (8)$$

$$SPE = \frac{TN}{TN + FP} \quad (9)$$

Have you noticed that recall (given by Eq. 1b) and sensitivity (given by Eq. 9) are the same things?

Problem1: **Compute the following performance measures for the balanced and unbalanced classes.**

Performance Measures	Balanced Data	Unbalanced Data
	TP=90 FN=10 TN=70 FP=30	TP=90 FN=10 TN=700 FP=300
SEN		
SPE		
ACC		
BACC		
PR		
RC (=SEN)		
F1 Score		
GM		
YI		
PPCR		

Problem 2: **Determine RFM scores of the customer based on the given imformation.**

Customer	Recency (day)	Frequency	Monetary (Total)
A1	9	19	189
A2	1	7	750
A3	25	18	706
A4	7	17	192
A5	30	20	909
A6	23	12	484
A7	3	17	938
A8	4	12	146
A9	22	15	678
A10	17	3	690

Solution guideline:

Customer	Recency (day)	Rank

Customer	Frequency	Rank

Customer	Moentary	Rank

Final RFM Score:

Customer	R	F	M
A1			
A2			
A3			
A4			
A5			
A6			
A7			
A8			
A9			
A10			